

ENERGY393T/695T: Energy Modeling of the Cape Town Energy System

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Executive Summary

Motivation: Cape Town and South Africa face an energy crisis as the result of a historically carbon intensive, coal-reliant, old and poorly maintained energy system. While the need for an energy transition is well known, aging infrastructure, mismanagement, and corruption within Eskom, South Africa's main utility, has resulted in systemic inefficiencies and inequities. The transition to clean and renewable fuel sources is pressing, but the overarching reliance on Eskom has made the city vulnerable to load shedding, showcasing a need for a more resilient energy system. Cape Town has already begun to see an increase in renewable energy generation, especially within the residential sector, through the adoption of small-scale embedded generation (SSEG). Despite this uptake, these additions are disproportionately pursued by high income households and large-scale businesses, having implications on energy injustice. While SSEG expansion promotes a more decentralized energy strategy, it is important to have regulatory oversight and tariff structures, that ensure economic and technical prosperity for all users during this transition.

Team Objective: This project aims to understand and address Cape Town's energy challenges to inform more equitable energy planning solutions; we construct an energy systems model that looks to mitigate load shedding and promote a more just energy transition.

Methodology: We used the Long-range Energy Alternatives Planning System (LEAP) for analysis. Historical data from 2018-2023 and expected capacity additions from South Africa's 2023 Integrated Resource Plan (IRP) inform the Business-As-Usual (BAU) scenario. Our City of Cape Town Renewable Energy Expansion (CCTREE) Scenario builds off the BAU adding more renewable energy (RE) capacity from SSEG. Our Conservative Integrated Resource Plan (CIRP) scenario assumes that Eskom builds out new supply slower than anticipated.

Demand and Supply model: Our demand modeling included a hybrid approach with bottom-up modeling of the residential sector, disaggregated by income group, and top-down modeling of other sectors differentiated by tariff including Large Power Users, Small Power Users, and the Municipality. The supply modeling analysis was broken down by ownership including Eskom, Eskom-contracted Independent Power Producers (IPP), SSEG/Distributed Generation, and Cape-Town-Owned Plants.

Results: For the BAU scenario, total electricity demand in Cape Town increases from 7,985.9 GWh in 2023 to 9,643.9 GWh in 2050 and installed generation capacity increases from 2,062.9 MW in 2023 to 5,216.0 MW in 2050. Capacity changes include the build out of utility-scale wind & solar, increased SSEG adoption, and coal retirements. The drop in coal generation, amidst growing demand and supply expansion, does not eliminate loadshedding. Our CCTREE scenario, which observes higher SSEG adoption, and our CIRP scenario, which includes lower SSEG adoption, convey that SSEG adoption inversely affects load shedding, especially if there are delays in new projects coming online.

Implications and Conclusions: Integrating SSEG into Cape Town's grid presents a critical opportunity to support clean energy goals and reduce load shedding. However, it also requires careful tariff reform to maintain financial stability and address equity concerns. Without tariff design changes, reduced electricity sales – especially from residential users – could erode municipal revenue. This may result in lower-income customers, who may not be able to afford SSEG, bearing a disproportionate share of rising grid costs.

Introduction and Motivation

South Africa stands at a crossroads in its energy and economic development journey, facing immense challenges that highlight the need for a nuanced and deliberate energy transition. This transition is more challenging due to global climate change, which demands collective action to limit global temperature increases to 1.5°C above pre-industrial levels. South Africa, as Africa's largest carbon emitter and responsible for approximately 40% of the continent's total greenhouse gas (GHG) emissions in 2022, plays a pivotal role in achieving this global goal. The country's electricity sector, which accounts for nearly half of its GHG emissions, is overwhelmingly reliant on coal, with 84% of electricity generation stemming from coal in 2022 (Country Commercial Guide, 2024). This heavy reliance not only intensifies environmental challenges but also exposes the inherent vulnerabilities in South Africa's energy system.

Cape Town, home to nearly five million people, exemplifies the complexities of South Africa's energy challenges. As the nation's largest metropolitan area, Cape Town is integral to advancing South Africa's energy transition. However, its energy landscape mirrors the broader national crisis: a carbon-intensive electricity supply, persistent load shedding due to inadequate generation capacity and aging transmission and distribution infrastructure, and inequitable access to reliable energy. These challenges are compounded by socioeconomic factors, including high levels of unemployment, poverty, and inequality, all of which shape the country's energy policy and complicate efforts to ensure a just energy transition.

The root causes of South Africa's energy crisis are multifaceted. Eskom, the state-owned utility responsible for 90% of the electricity supply, has been plagued by aging infrastructure, financial instability, and decades of mismanagement (Eskom, Generation Plant Mix, 2024). Frequent breakdowns at coal-fired power plants have resulted in a persistent energy undersupply, prompting load shedding that has occurred regularly since 2007. Encouragingly, load shedding has been minimal in 2024, with no scheduled outages reported since March. Nevertheless, the energy supply situation remains fragile (Cowling, 2024). These outages disrupt economic activity, harm public health, and exacerbate environmental issues as individuals and businesses turn to diesel generators during power cuts. Over time, the problem has been exacerbated by mismanagement of the national grid, corruption within utilities, high levels of debt, and failed investments. A notable example is the \$3 billion World Bank loan in 2010, which financed the Medupi coal-fired power plant. Although Medupi is now operational, the project faced significant delays and cost overruns, highlighting systemic inefficiencies. Furthermore, Eskom's significant debt burden—driven by mismanagement, delays, cost overruns, and operational inefficiencies—continues to impede its capacity to modernize infrastructure, thereby perpetuating recurring outages and financial instability (Baker, Burton, Godinho, & Trollip, 2015).

The global push for decarbonization adds another layer of urgency. South Africa's coal-dependent energy model is one of the most carbon-intensive in the world, making it a focal point for international climate scrutiny. The impacts of climate change such as rising sea levels, severe flooding, heatwaves, and prolonged droughts are already being felt, underscoring the need for a swift transition to a resilient and sustainable energy system. South Africa's Nationally Determined Contributions (NDCs) aim to reduce emissions to 350-420 mega tonnes of CO₂ by 2030. The Just Energy Transition Investment Plan (JET IP) implemented in partnership between South Africa and the International Partners Group (IPG)—which includes France, Germany, the United Kingdom, and the European Union—aims to support South Africa's energy transition, with an initial pledge

of \$8.5 billion. The partnership emphasizes a "just transition" that balances emissions reductions with socioeconomic equity, addressing the direct and indirect impacts of the transition on workers, communities, and livelihoods. The formation of JETP at COP26 presents a hopeful prospect for South Africa, marking a significant step towards a sustainable energy future. (South Africa's Just Energy Transition Investment Plan (JET IP), 2022).

Cape Town's energy transition faces unique hurdles. The city's reliance on Eskom makes it vulnerable to load shedding and underscores the need for decentralized, resilient energy systems. While the adoption of small-scale embedded generation (SSEG) such as solar photovoltaic (PV) systems has grown, especially among affluent households and businesses, this uptake has highlighted the disparities in energy access. High-income groups benefit disproportionately from decentralized solutions, leaving marginalized communities further behind. Addressing these disparities requires deliberate efforts to ensure equitable access to renewable energy technologies and the associated benefits. Policy frameworks such as the Integrated Development Plan (IDP) 2022-2027 and the National Development Plan: Vision 2030 set ambitious targets for Cape Town, including carbon neutrality by 2050. Achieving these goals will necessitate significant reforms, such as the unbundling of Eskom to allow for competition and market-driven efficiency, and the establishment of an Independent System Operator (ISO) to manage the grid. Additionally, targeted investments in renewable energy infrastructure, feed-in tariffs, and community-owned energy projects will be critical to fostering inclusivity and resilience (National Development Plan 2030).

This project focuses on addressing Cape Town's energy challenges by developing modeling and analysis to inform equitable and efficient solutions. By leveraging tools such as the Long-range Energy Alternatives Planning System (LEAP), the study aims to evaluate various scenarios for mitigating load shedding and promoting a just energy transition. The insights generated will contribute to inclusive renewable energy policies and ensure that the transition aligns with South Africa's developmental and climate goals.

Relevant Literature

South Africa's Energy Transition

The growth of renewable energy technologies has opened opportunities for decentralized electricity production, allowing consumers to become "prosumers"—both producers and consumers of electricity. This innovation can help reduce emissions while building a more resilient grid ensuring an optimal solution to South Africa's energy challenges (Goldthau 2014). Despite this, ensuring an inclusive, equitable, and sustainable energy transition is challenging. This task is especially complicated in the Global South, where climate change impacts are expected to be severe, social inequalities are deeply rooted, and the demand for economic growth is urgent. SSEG presents a promising pathway for decentralized and low-carbon energy generation (Baker and Phillips 2018). This approach enables households to generate their own electricity, reducing reliance on the grid and potentially lowering electricity costs, especially during loadshedding events. However, SSEG installations are accessible primarily to higher-income households that can afford the significant upfront investment and often costly maintenance associated with solar photovoltaic (PV) and battery systems.

Cape Town Specific Considerations

The City of Cape Town charges a monthly fee for an SSEG grid connection, so there is a notable level of non-compliance regarding system registration. A study by Kritzing et al. 2020 found that 50% of Cape Town homeowners with SSEG systems had not registered. In consequence, the city and utility miss out on potential profits while also having inaccurate estimates of the city's electricity capacity resulting in poor energy planning. This challenge is unique because the City of Cape Town has the responsibility to maintain municipal revenues from electricity sales to fund critical services for the broader population, but greater SSEG adoption could allow for increased energy independence and resiliency for households (Baker and Phillips 2018). Different incentive structures have been shown to influence the level of SSEG adoption and even the generation capacity homeowners invest in, affecting overall electricity supply and inadvertently municipal revenue as SSEG adoption decreases the amount of electricity demanded from the City (Lee 2017).

The rise of SSEG in Cape Town reflects broader socioeconomic divides in energy access, allowing high-income households to “opt-out” of the municipal grid (either by not being connected to the grid at all or being connected but not registering their SSEG system) and avoid escalating tariffs, while low-income consumers remain reliant on an increasingly costly and unreliable coal-fired electricity system (Baker et al 2015). Furthermore, since SSEG investments are clustered in high-income neighborhoods, municipalities are spending more time and money on upgrading grid infrastructure in those areas to accommodate the solar connections, despite the fact they will receive lower revenues from those neighborhoods going forward (Oliver 2021). This leads to growing disparities in both physical network infrastructure between high- and low-income neighborhoods and electricity costs, with homeowners without PV systems bearing a disproportionate burden of subsidizing low-income household electricity expenses.

Filling the Research Gap

A Mayr et al. 2015 paper discusses a least cost optimization model of electricity in Cape Town, which shows that increased embedded generation reduces municipal revenues. They construct Cape Town's demand profile by dividing households based on socioeconomic status and electricity consumption, and each group is assigned hourly load profiles representing typical demand, which inspires demand construction in our model. This paper mentions that the then recently instituted feed-in tariff would likely affect SSEG adoption and municipal revenues, but the tariff wasn't included because of uncertainty on how it would change over time (Mayr et al., 2015). Future work may incorporate the feed-in tariff to better understand affordability with increased SSEG adoption.

The Energy Systems Research Group (ESRG) based out of Cape Town University has used the SATIMGE modeling framework to understand the economic and development tradeoffs of a low carbon transition in South Africa. The SATIMGE framework links the South African TIMES model, a bottom-up energy systems model, to eSAGE, a computable general equilibrium model, running iteratively and passing inputs to each other at each time step.

In 2015, updated Energy Scenarios for Cape Town was released to identify energy use and emissions trends that could inform policy and future modeling. The report outlined using LEAP to model various electricity and transport efficiency policies, supply side capacity additions, and sensitivity analyses (City of Cape Town & SEA, 2015). Also in 2015, Cape Town released an

updated State of Energy report that used LEAP to investigate an increase in SSEG adoption. The analysis concluded that increased SSEG resulted in lower emissions and lower costs compared to the business-as-usual scenario. However, the LEAP model did not indicate if it was the households, the municipality or the utility that benefited from those cost savings.

Our research builds off these three studies, combining the use of LEAP, focusing on the socioeconomic impacts of a low-carbon transition in Cape Town. Specifically, the socioeconomic impacts of increased SSEG adoption and load shedding. ESRG and the State of Energy and Energy Futures studies both incorporate capacity additions from South Africa's Integrated Resource Plan, which we also incorporated into our methodology.

Methodology

In the Energy Modeling workstream, we incorporated distributed solar generation potential into a Cape Town energy system model to evaluate the implications for meeting energy demand while supporting the municipality's targets to deliver an affordable, clean, and reliable electricity supply. We utilize the LEAP model, a widely used tool for energy policy analysis and climate change mitigation assessment, to undertake quantitative modeling of Cape Town's energy system (Stockholm Environment Institute, n.d.-b). LEAP is particularly suitable for our analysis as it has already been used to inform energy planning in Cape Town and can be used for both full energy system optimization modeling and energy affordability analysis (Stockholm Environment Institute, n.d.-a).

LEAP model

Our team compiled technical and economic data to input into the LEAP model, largely provided by the City of Cape Town, alongside other publicly available datasets used to develop South Africa's latest Integrated Resource Plan (IRP). Our team also compiled economic information, including PV system and battery storage costs, electricity cost data, and tariff information.

Energy Demand Model

LEAP supports several demand side modeling methodologies including bottom-up, end-use accounting techniques and top-down macroeconomic modeling (Heaps, n.d.). To represent energy demand for our analysis, we utilized a hybrid approach with a top-down representation of baseline electricity demand (*Introductory LEAP Training Course*, 2022). The focus of our analysis is final energy consumption, specifically electricity consumption, which we disaggregated by residential households, small power users (SPUs), large power users (LPUs), and the Municipality. We further divided the residential households into electrified and non-electrified, with each sub-division further divided into low income, middle income, and high income. Among each income group, we classified homes into two categories: those supplied by grid electricity only and those supplied by a combination of electricity from the grid and SSEG.

We utilized the branch structure in LEAP to define the disaggregation and specify the final energy consumption in special branches called "technologies" (*Introductory LEAP Training Course*, 2022). Four types of technologies are available in LEAP including "*Technology with Energy Intensity*", "*Technology with Total Energy*", "*Transport Technology*" and "*Other Technology*".

We utilized the first two of these methodological options for our analysis in LEAP. Final energy consumption in the residential sector was defined using the “*Technology with Energy Intensity*” methodology, where energy consumption is represented as:

$$\text{Energy Consumption } (EC_{s,t}) = \text{Activity Level } (AL_{s,t}) \times \text{Energy Intensity } (EI_{s,t})$$

where s is the scenario evaluated, and t is the simulation year. The Activity Level is represented by the number of households of each type, and the energy intensity by the corresponding average energy consumption per household category. Inputs for the rest of the consumer categories, including small power users (SPUs), large power users (LPUs), and the municipality, were represented using the “*Technology with Total Energy*” methodology, using the value of total final electricity consumption for the respective sectors.

Household Branch

Approach to Modeling the Activity Level

The primary dataset used for defining energy demand in the LEAP model was electricity sales summary records provided by the City of Cape Town. The electricity sales dataset disaggregates consumption by tariff, listing the number of consumers billed and energy consumption both monthly and annually. The annual values are summarized in worksheets named ‘*Summated Sales*’, available for each year where electricity sales summary data was provided from the 2012-2013 financial year to the 2022-2023 financial year. We utilized these annual values of the number of consumers in each residential tariff category to define the number of households under each tariff, assuming one residential consumer represents one household. The aggregated tariff categories for residential consumers include Domestic, Home User, Lifeline, and Residential SSEG. For our analysis, we combined “Residential SSEG” consumers with those under the Home User tariff. This decision was made as the City of Cape Town lists the tariff as available only for approved SSEG connections in conjunction with the Home User tariff (City of Cape Town, 2022).

Table 1. Number of households estimated from the number of consumers in each tariff category

Tariff Category	17/18	18/19	19/20	20/21	21/22	22/23
Lifeline	197,194	184,716	168,108	163,335	164,841	172,697
Domestic	177,631	175,075	159,865	157,352	156,798	150,534
Home User ¹	207,538	236,181	277,539	277,225	287,789	310,550
¹ For our analysis, consumers under the Residential SSEG tariff category are included in the Home User tariff						

As the residential electricity sales summary data from the City of Cape Town was categorized by tariff, but our LEAP model was categorized by income, we estimated the proportion of consumers (or households) from each tariff category in each income group using demographic data from the City of Cape Town State of Energy and Carbon (SOEC) Report alongside residential property market data from the Centre for Affordable Housing Finance (CAHF) in Africa (CAHF, 2021).

Residential tariff information from the City of Cape Town, summarized in Table 2, highlights several factors that impact the tariff for residential consumers. These factors include 12-month average electricity usage, property value, meter type, and household rebate registration. We used the property values from the tariff classification summarized in Table 2, combined with the data from the CAHF 2021 Housing Market Report, to allocate households in each tariff category to income groups.

Table 2. *City of Cape Town Tariff Group Economic Information*

Tariff Group	Block	c/kWh	c/kWh (plus VAT)	Usage (kWh)	Constraints
Lifeline	1	206.16	237.08	0-600	Property value of ZAR500,000 or less and receives less than 450 kWh or less prepayment meter
Lifeline	2	206.16	237.08	600+	
Lifeline Free Electricity	1	NA	NA	60	If you receive less than 250kWh per month on average taken over the previous 12-month assessment period,
Lifeline Free Electricity	1	NA	NA	25	If you receive more than 250 kWh but less than 450 kWh per month on average taken over the previous 12-month assessment period
Domestic	1	339.74	390.7	0-600	450 kWh+ Usage and property value above ZAR 500,000
Domestic	2	413.38	475.49	600+	
Home User	1	298.7	343.51	0-600	ZAR1,000,000 property value or more
Home User	2	413.38	475.39	600+	

Sources: (City of Cape Town 2024) (Electricity Generation and Distribution Department, n.d.)

Number of residential properties by market segment
City of Cape Town, 2021

NB: 2021 figures are as at 30 June 2021.

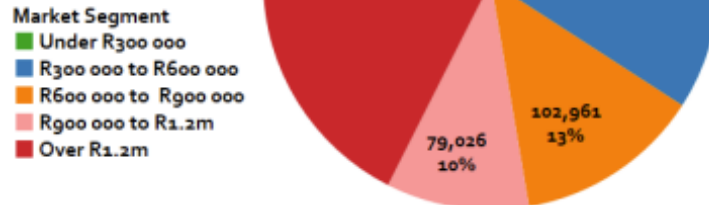


Figure 1: Number of residential properties by market segment from the CAHF 2021 Cape Town Housing Market Report ([Source](#))

The CAHF 2021 Housing Market Report uses title deeds data to classify residential properties in Cape Town into five segments according to their value, as illustrated in Figure 1 and summarized in Table 3. Since the dataset only includes properties in the registry, the numbers provided only include properties in the formal market and exclude informal settlements (CAHF, 2021).

Table 3. Proportion of residential properties by value from the CAHF 2021 Housing Market Report

Category	Property Value	Quantity	Percentage
Entry market	R300,000 or less	128,349	16%
Affordable market	R300,000 – R600,000	138,075	18%
Conventional market	R600,000 – R900,000	102,961	13%
High-end market	R900,000 – R1.2 million	79,026	10%
Luxury market	Properties over R1.2 million	331,852	43%

To assign the households to income categories using the three datasets, we followed the steps below:

1. We used a Python script and the percentages of properties in each category summarized in Table 3 to generate random lists of property values proportional to the number of properties in each category. Specifically, we generated a list of 1600 random numbers (0-300,000), 1800 random numbers (300,000-600,000), and so on until we had a list of 10,000 properties with a distribution like that in Table 3.
2. Using the distribution of property values randomly generated in Step 1, we calculated the percentage of properties with a value less than R500,000, values between R500,000 and R1 million, and values above R1 million in line with the property classification used in the tariff description in Table 2.

Table 4. Proportion of households in each tariff category from the randomly generated dataset of property values

Proportion of households in each tariff category	Percentage
Lifeline	28.0%
Domestic	22.4%
Home User	49.6%

Based on the data from Cape Town’s State of Energy and Carbon Report, the proportion of low-income, middle-income, and high-income households in the City of Cape Town was estimated to be 48%, 27%, and 26%, respectively. We used these percentages alongside the data in Table 4 to estimate the percentage of low-income, middle-income, and high-income households (HHs) on the Lifeline and Domestic tariffs respectively, as follows:

Low Income (LI)

$$\% \text{ of LI HHs on Lifeline tariff} = \frac{\% \text{ of HHs on Lifeline tariff}}{\% \text{ of LI HHs}}$$

$$\% \text{ of LI HHs on Domestic tariff} = \frac{\% \text{ of LI HHs} - \% \text{ of HHs on Lifeline tariff}}{\% \text{ of LI HHs}}$$

Middle Income (MI)

$$\% \text{ of MI HHs on Domestic tariff} = \frac{\% \text{ of HHs on Domestic tariff} - (\% \text{ of LI HHs} - \% \text{ of HHs on Lifeline tariff})}{\% \text{ of MI HHs}}$$

$$\% \text{ of MI HHs on Home User tariff} = \frac{\% \text{ of MI HHs} - (\% \text{ of HHs on Domestic tariff} - (\% \text{ of LI HHs} - \% \text{ of HHs on Lifeline tariff}))}{\% \text{ of MI HHs}}$$

High Income (HI)

All high-income households are assumed to be on the Home User tariff.

Based on the percentages calculated above, the distribution of the households to the income groups was implemented as summarized in Table 5.

Table 5. Distribution of households by income groups and tariff category

A Income Group	B Total No. of Households ¹	C Relative Proportion ¹	D Number of HHs in each tariff per income group ²		E Estimated total no. of HH per tariff category in each income group ³
High Income	306,898	26%	Home User	100%	306,898.00
Middle Income	322,513	27%	Domestic	9%	29,026.00
			Home User	91%	293,487.00
Low Income	571,755	48%	Lifeline	58%	331,618.00
			Domestic	42%	240,137.00

¹We sampled the total number of households and relative proportions in each income group from the City of Cape Town 2021 State of Energy and Carbon Report

²We used the equations described above to estimate the number of households in each tariff category based on the income categories (Column D)

³Using the values calculated in Column D, we estimated the total number of households per tariff category in each income group by applying the percentages in Column D to the sampled number of households in each income group in Column B

Using the values in Column D above, we estimated the relative proportions of households in each tariff that were either low income, middle income and high income by taking the total number of consumers in each tariff category as classified in Column D and calculating the percentages estimated to be allocated to each income group in Column A. We then applied the estimated proportions from these calculations, summarized in Table 6, to the number of households from the City of Cape Town dataset in Table 1 to compute the numbers that were input to the LEAP model. The resulting number of households per income group used for our analysis is summarized in Table 7.

Table 6. Relative proportions of households in each tariff estimated to be in the low income, middle income and high-income categories

Proportion of consumers in each tariff in each income group:	Low Income	Middle Income	High Income
Lifeline	100%	0%	0%
Domestic	89%	11%	0%
Home User	0%	49%	51%

Table 7. Estimated number of households in each income group calculated using the percentages in Table 6

Income Grouping	17/18	18/19	19/20	20/21	21/22	22/23
Low Income	355,670	340,911	310,733	303,718	304,730	306,998
Middle Income	120,606	134,332	152,909	152,485	157,589	168,040
High Income	106,087	120,728	141,869	141,709	147,109	158,743

Approach to Modeling Energy Intensity

The same percentages, defined in Table 6, used to apportion the households in each tariff category to income groups, were also used to apportion the billed energy consumption across the income groups. The energy intensity, in kWh per household, was then calculated by dividing the total energy consumption apportioned to an income group by the total number of households estimated to belong to the same income group based on similar proportions.

Approach to Modeling Load Shedding Impacts

To evaluate load shedding, which is represented by unmet requirements in LEAP, we needed to estimate the demand that would have occurred but was not met due to load shedding. Our current demand figures are based on electricity sales summaries, which account for actual demand met and don't account for subdued demand due to load shedding. To incorporate load shedding, we reference data from Eskom's annual integrated reports, which estimate the percentage of demand not met due to load shedding and load curtailment, estimated to be 5.92% in 2024 (Eskom, 2024). We applied these annual percentages to historical data as a factor in our calculated demand figures from the electricity sales summaries.

Approach to Modeling Grid vs Solar Consumption

As highlighted in the description of our energy demand model and illustrated in our model overview, we classified homes into two categories: those supplied by grid electricity only and those supplied by a combination of electricity from the grid and SSEG, also known as prosumers. To estimate the proportion of households in each income group with SSEG, we used data from the General Household Survey 2023. We took the midpoint of the past month's household expense buckets and divided them by the number of people in each household to calculate a proxy for income. We then stratified household income level so that the proportions are equivalent to demographics in the State of Carbon and Energy Report. However, in order to not artificially break ties, the resulting income partitions were 26.05%, 25.26%, and 48.68% for high, middle, and low income, respectively.

From our survey data analysis, we estimated that 2.46% of high-income households have SSEG while 0.09% of middle-income households also have SSEG. For our initial analysis, we assumed that low-income households are not consuming energy from SSEG. The step-by-step iterative process of estimating household consumption from SSEG was done as follows:

1. By multiplying 2.46% by the number of households categorized as high Income, we estimated the number of high-income households with SSEG. Using the baseline high income household energy intensity calculated above, we began with a baseline allocation of 15% of the consumption to SSEG and estimated an initial value of consumption by these households (i.e. No. of High Income (HI) HHs $\times$ % of HI HHs with SSEG $\times$ Average Energy Consumption by HI HH $\times$ 15%). The baseline allocation of 15% was based on a 2024 RES4Africa report that assessed techno-economic potential of rooftop PV in Cape Town, estimating the proportion of consumption that could be met by a solar home system under different grid conditions (Riva et al., 2024).
2. We repeated the same procedure for middle income households. By multiplying 0.09% by the number of households categorized as middle income, we estimated the number of middle-income households with SSEG. Using the baseline middle income household energy intensity, we began with a baseline allocation of 21% of the consumption to SSEG and estimated an initial value of consumption by these households (i.e. No. of Middle Income (MI) HHs $\times$ % of MI HHs with SSEG $\times$ Average Energy Consumption by MI HH $\times$ 21%). (Note: Initial percentages are based on the RES4Africa report.)
3. Using the figures in Steps 1. and 2. above, we arrived at total consumption from SSEG, assuming low income SSEG consumption to be zero for our baseline analysis. However, since the calculation above was based on the survey data, we noted there would be a difference with the total residential consumption from SSEG, estimated from the amount of residential SSEG capacity in our supply model. To address this, we used a goal seek to find a value for the percentage of SSEG energy consumption by high income households (the 85:15 mix) until the total from our estimates based on Steps 1. and 2. above matched the total generation we estimated from installed residential SSEG capacity in 2018.
4. For subsequent years in the Current Accounts (2019 – 2023), we kept the percentage of energy consumption met by SSEG for high income households calculated in Step 3. above constant. However, since the installed residential SSEG capacity is increasing year on year, we needed to reconcile this growth on the demand model by increasing the proportion of households with SSEG. To reconcile the totals, we used a goal seek to adjust the 2.46% figure used in Step 1. above upwards until the totals from the SSEG generation in the supply model and Step 3. above matched.

An overview of the electricity demand model for the BAU scenario created using the inputs described above, for the Base Year, 2018, is shown in Table 8.

Table 8. Annual household consumption in 2018 by income level in kWh and Grid/Solar Home System Status

Current Accounts Demand Parameters	
Households	
582,363	
Electrified	
100%	
582,363	

Current Accounts Demand Parameters								
High Income			Middle Income			Low Income		
18.22%			20.71%			61.07%		
106,107			120,607			355,649		
Grid Only	Grid and SHS		Grid Only	Grid and SHS		Grid Only	Grid and SHS	
18.22%	20.71%		61.07%	18.22%		20.71%	61.07%	
106,107	120,607		355,649	106,107		120,607	355,649	
Grid Only Cons.	Grid Cons.	SHS Cons.	Grid Only Cons.	Grid Cons.	SHS Cons.	Grid Only Cons.	Grid Cons.	SHS Cons.
6,532.74	6,532.74		6,346.67	6,346.67		3,887.05	3,887.05	
100%	84%	16%	100%	79%	21%	100%	100%	0%
6,532.74	5,491.84	1,040.90	6,346.67	5,020.22	1,326.45	3,887.05	3,887.05	0.00

LPU, SPU and Municipality Branches

As highlighted in the description of the energy demand model, energy consumption for LPUs, SPUs and the Municipality was defined using total energy across all consumers grouped under these categories. Like households, we defined a share of the energy consumption from these consumer categories that was met by SSEG. To do so, we estimated the total non-residential generation, based on installed SSEG capacity records from the City of Cape Town, using a similar capacity factor for SSEG as that in the 2021 State of Energy and Carbon Report. Using the share of SSEG generation supplied to the City of Cape Town by each tariff category from the feed-in records, we apportioned the estimated annual generation in each year to LPUs and SPUs, subsequently adding this SSEG generation to total grid consumption by consumers under each tariff category.

Table 9. *Historical annual consumption (2018-2023) of LPU, SPU and Municipality sectors in GWh/year (City of Cape Town Electricity Sales Summaries)*

Classification	2018	2019	2020	2021	2022	2023
Large Power User (LPU)	3648	3567	3080	3323	3284	3047
Small Power User (SPU)	1694	1697	1478	1531	1534	1393
Municipality	351	334	322	322	311	299

Approach to Modeling Future Energy Demand Values

We modeled changes in energy demand using the projected South Africa wide compounded annual growth rates (CAGRs) for energy demand per sector from the 2023 Integrated Resource Plan (IRP) (Merven & Tatham, 2023). The energy demand projections, developed using the linked energy-economic model for South Africa (SATIMGE), specify growth rates of 0.7%, 0.9% and 0.1% for the residential, manufacturing, and commercial sectors respectively (Merven, 2023).

Since residential demand in our model is broken down by households and energy intensity, we modified both the household population growth and annual changes in energy intensity per household to achieve the total demand growth rate of 0.7%. To achieve this, we calculated the projected CAGR of the number of households in the City of Cape Town, which estimates that the number of households will increase from 1.5 million in 2024 to 2.1 million in 2050 (Simpson, 2025). If we applied the resulting estimated CAGR of 1.30% to the number of households in our model, then our estimated energy consumption would also increase by 1.30% annually until 2050, well over the projected annual increase in total residential energy demand in the IRP. To correct for this, we adjusted the CAGR of the household energy intensity until the total growth in residential energy demand matched the 0.7% in the IRP, resulting in a final estimated growth rate of -0.6% applied to household electricity consumption. While the final growth rate for household energy intensity turns out to be negative, this assumption is plausible due to energy efficiency improvements and fuel switching offsetting the population growth, as highlighted in the demand projection model report (Merven & Tatham, 2023). Due to limited data, our model also adopts a simplified approach by assuming no changes in the relative proportion of households across income groups and changes in energy consumption within those income groups.

Since electricity demand for LPUs, SPUs, and the Municipality as total energy, we directly applied the annual growth rates from the IRP of 0.9%, 0.1%, and 0.1% respectively for these demand branches. For simplicity, we assumed that LPUs are predominantly from the Manufacturing sector while the SPUs and Municipality would fall under the Commercial category in the IRP (Merven, 2023).

Daily Demand Load Curve by Sector

We used sector-specific demand load curves to account for seasonal variation, sourcing data from the City of Cape Town for the Household/Residential, LPU, and SPU sectors. Municipal load curves were not specified due to limited data.

We cleaned and aggregated sector-specific load interval data into month and hour distributions to import into LEAP. This resulted in 288 total input values, or “time slices,” across the twelve months of a year. Peaks can be seen in the months of July and August, characteristic of high solar water heater usage in the winter months.

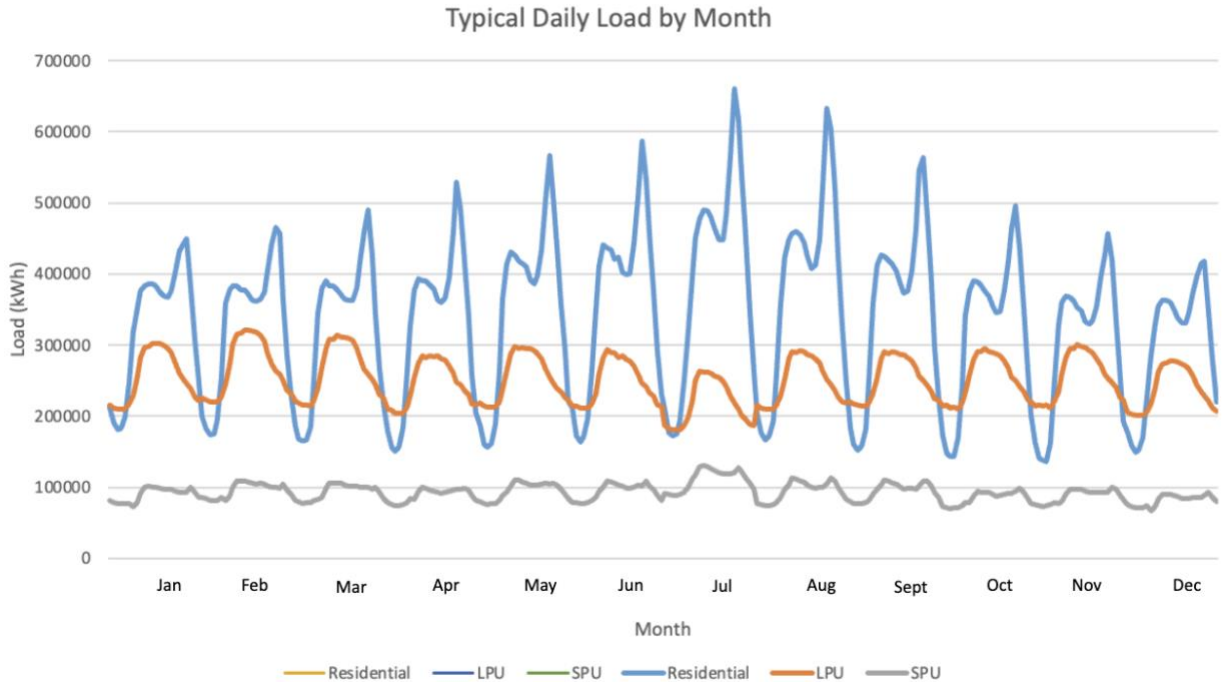


Figure 2: Typical daily load curve by month imported into LEAP

Energy Supply Model

Electricity generation in LEAP was defined using the Transformation and Resources modules in LEAP. This was done by adding grid-connected power plants, scaled to a representative capacity, to meet the city's electricity demand. The three main components under Transformation and Resources were Electricity Generation, Distributed Generation, and Transmission & Distribution Losses. We selected the base year as 2018 to correspond with IRP assumptions. For 2018-2023, data was input in respective components, including exogenous capacity, historical production, lifetime, maximum availability, variable costs, and losses. The data came from multiple sources, and many assumptions were also made. Below is an overview of all assumptions and data used:

1. LEAP module - Electricity Generation:

a. Eskom Generation & IPPs

We utilized Eskom Integrated Report Historical Data (Integrated results, 2021) to understand its fuel mix, including the purchases from independent power producers (IPP) that it does. Eskom plants consist of Coal, Nuclear, Open-cycle gas turbines (OCGT), Hydro, and Wind. It also purchases electricity from other IPPs, which includes a supply mix of Wind, Solar, and OCGT. All minor other renewable sources were clubbed together as 'Other Renewables', and smaller non-renewable sources were clubbed together as 'Other Non-Renewables'. After defining the capacity (MW) of Eskom-owned entities, we calculated the percentage of supply mix from each source (% breakup from capacity) and worked to scale the capacity down to what Cape Town's capacity would look like. This was done by finding Cape Town's peak demand for Eskom purchases. The peak demand figures were found from Cape Town government committee meeting minutes (Gov.,

n.d.) for the years 2018-2023. After peak demand and the percentages of the supply mix were found for Eskom plants & IPPs, these values were multiplied together to get the exogenous capacity (MW) for Cape Town coming from Eskom plants.

Table 10: Eskom own and IPP plants Capacity (MW)

	2018	2019	2020	2021	2022	2023
Peak Demand MW	1,759	1,782	1,715	1,719	1,701	1,805
<i>Breakup:</i>						
<u>Eskom – Own plants:</u>						
Coal	1,323	1,322	1,276	1,268	1,243	1,309
Nuclear	65	67	63	61	58	62
Hydro	116	121	113	109	105	111
OCGT	84	87	82	79	76	81
Wind	3	4	3	3	3	3
BESS	-	-	-	-	-	-
<u>Eskom – IPP plants:</u>						
Wind	69	72	67	78	97	112
Solar	62	72	75	87	85	91
OCGT	35	36	34	33	32	34
Other RE	1	1	1	1	2	2
Other - Non-RE	-	-	-	-	-	-
BESS	-	-	-	-	-	-

The historical production (GWh) came from multiplying the recorded total energy imports from Eskom based on the data provided by the City of Cape Town (Energy Parameters Excel file: ‘Energy Balance’ sheet) with the breakup of % capacity by source highlighted in Table 10. The maximum availability (in this case: capacity factor) was determined by the LEAP model using capacity vs historical production through the formula:

$$\rightarrow \text{Production (GWh)} = \text{Capacity (MW)} \times \text{Capacity Factor (\%)} \times \text{Hours in a year}$$

Table 11: Eskom-own and IPP plants Production (GWh)

	2018	2019	2020	2021	2022	2023
Eskom Imports (GWh)	9,525	9,520	8,988	9,138	8,953	8,264
<i>Breakup:</i>						
<u>Eskom – Own Plants:</u>						
Coal	8,276	8,270	7,688	7,764	7,411	6,755
Nuclear	581	478	524	419	496	387
Hydro	212	232	227	262	268	282
OCGT	5	50	53	62	73	119

	2018	2019	2020	2021	2022	2023
Wind	14	14	11	13	10	8
<u>Eskom – IPP plants:</u>						
Wind	260	262	251	325	394	402
Solar	165	185	198	256	248	254
OCGT	4	22	28	30	36	43
Other RE	7	7	7	9	17	14
Other - Non RE	-	-	-	-	-	-

Table 12: Eskom-own and IPP plants Capacity Factors %:

	2018	2019	2020	2021	2022	2023
<u>Eskom – Own Plants:</u>						
Coal	61%	63%	59%	54%	53%	50%
Nuclear	87%	71%	81%	61%	76%	60%
Hydro	18%	19%	20%	21%	23%	25%
OCGT	1%	6%	6%	7%	9%	14%
Wind	38%	37%	32%	35%	29%	24%
<u>Eskom – IPP plants:</u>						
Wind	37%	37%	37%	37%	37%	35%
Solar	26%	26%	26%	26%	26%	27%
OCGT	1%	6%	8%	8%	10%	12%
Other - Non RE	0%	0%	0%	0%	0%	0%

Lifetime was assumed to be 30 years for all plants. Variable costs mainly came from fuel costs of the different plants. Fuel cost assumptions were from EPRI's 2021 report as it reflected as most accurate for plants in South Africa (EPRI, 2021). The variable costs for Other Renewables were taken as an average of all renewable sources, and Other Non-Renewable were taken as an average of all non-renewable sources.

Table 13: Variable Costs as per EPRI Report 2021

	Fuel cost (ZAR/MWh)
Coal	374
Nuclear	119
Gas Peaking	1,355
Wind Onshore	-
Wind Offshore	-
Solar PV	-
Res Solar	-
Hydro	-

	Fuel cost (ZAR/MWh)
Biogas / biopower	2
Landfill	2
Other RE	1
Other Non RE	865

b. Cape Town Generation & IPPs

We utilized Cape Town's SOEC report (State of Energy and Carbon, 2021) to inform purchases from independent power producers (IPP). The City of Cape Town has two gas turbines and one storage plant: Athlone Gas Turbine (AGT), Rogge-Baai Gas Turbine (RGT), and Steenbras Pumped Storage (SPS). The gas turbines were decommissioned 20 years ago but are still used in emergencies to mitigate load shedding. The data from this report was used to find exogenous capacity (MW) for all Cape Town generation. We also found data on Cape Town purchases from IPPs, including from Darling Wind Power and micro hydro; however, we did not use this data in our model due to inconsistencies between capacity and IPP purchase information.

Table 14: Cape Town plants Capacity (MW)

	2018	2019	2020	2021	2022	2023
Steenbras Pumped Storage (SPS)	180	180	180	180	180	180
Athlone Gas Turbine (AGT)	36	36	36	36	36	36
Rogge-Baai Gas Turbine (RGT)	42	42	42	42	42	42

The historical production, which was reported in GWh, came from Cape Town's Open Data Portal (Portal, 2025) and we verified the production figures with the SOEC report 2021 to ensure consistency. Production figures were not reported directly from the SOEC report due to limited data availability. In the historical production figures we found, there were many negative figures for the SPS facility. This may be due to pumped-storage hydro systems consuming more electricity to pump water uphill, compared to the electricity they generate, resulting in a net energy loss (EIA, 2023). To ensure production didn't include negative values, zero generation was assumed for SPS.

Table 15: Cape Town plants Production (GWh)

	2018	2019	2020	2021	2022	2023
Steenbras Pumped Storage (SPS)	8	-	-	-	-	-
Athlone Gas Turbine (AGT)	0	0	1	0	1	-
Rogge-Baai Gas Turbine (RGT)	0	0	0	0	0	0

The maximum availability (in this case: capacity factor) for the plants owned by the City of Cape Town was determined by the LEAP model using capacity vs historical production through the formula shared in the Eskom section. Most capacity factors were assumed as 0% due to minimal production. Lifetime was defined as 30 years for all plants. The variable costs came from the fuel costs of the different plants. For fuel costs, we took the data from the EPRI 2021 report, as it reflected the most accurate costs for plants in South Africa (EPRI, 2021).

2. LEAP Module - Distributed Generation:

The distributed generation was determined by the SSEG profile in Cape Town. This is basically the SSEG feed-in to the Cape Town grid. We utilized the data provided by the City of Cape Town on SSEG contracts for past years (Energy Parameters Excel file: 'SSEG_Summary' sheet). We filtered out for only the contracts representative of installed SSEG highlighted below, totaled the generator capacity (VA), and converted it to total capacity (MW), which includes residential and commercial:

- Approved grid-tied installation commissioned
- Approved off-grid installation
- Approved grid-tied installation commissioned (legalized)
- Approved Standby / Passive UPS / Alternative supply installation
- Approved grid-tied installation decommissioned

The historical production (GWh) fed into the grid was sourced from the electricity sales data provided by the City of Cape Town (Energy Parameters Excel file: 'Energy Balance' sheet). The maximum availability was determined to be 19.6% as per Cape Town's SOEC 2021. If we let LEAP calculate the maximum availability based on only energy fed into the grid, then it would be set too low, as historical capacity factor figures based on the above capacity and production show that not much was used from SSEG. However, actual availability of SSEG is more than that, and using the value of 19.6% ensures future generation is representative of the total generation from SSEG, and not only that which is fed into the grid.

Table 16: SSEG feed-in data

	2018	2019	2020	2021	2022	2023
Capacity (MW)	19	31	50	73	99	121
Production (GWh)	1	4	8	10	13	17
Capacity Factor %	19.60%					

3. LEAP Module – Transmission & Distribution Losses:

The transmission & distribution losses branch only applies to electricity generation and not distributed generation. The losses were calculated using the formula below, based on total supply and total sales figures obtained from the data provided by the City of Cape Town (Energy Parameters Excel file: 'Energy Balance' sheet).

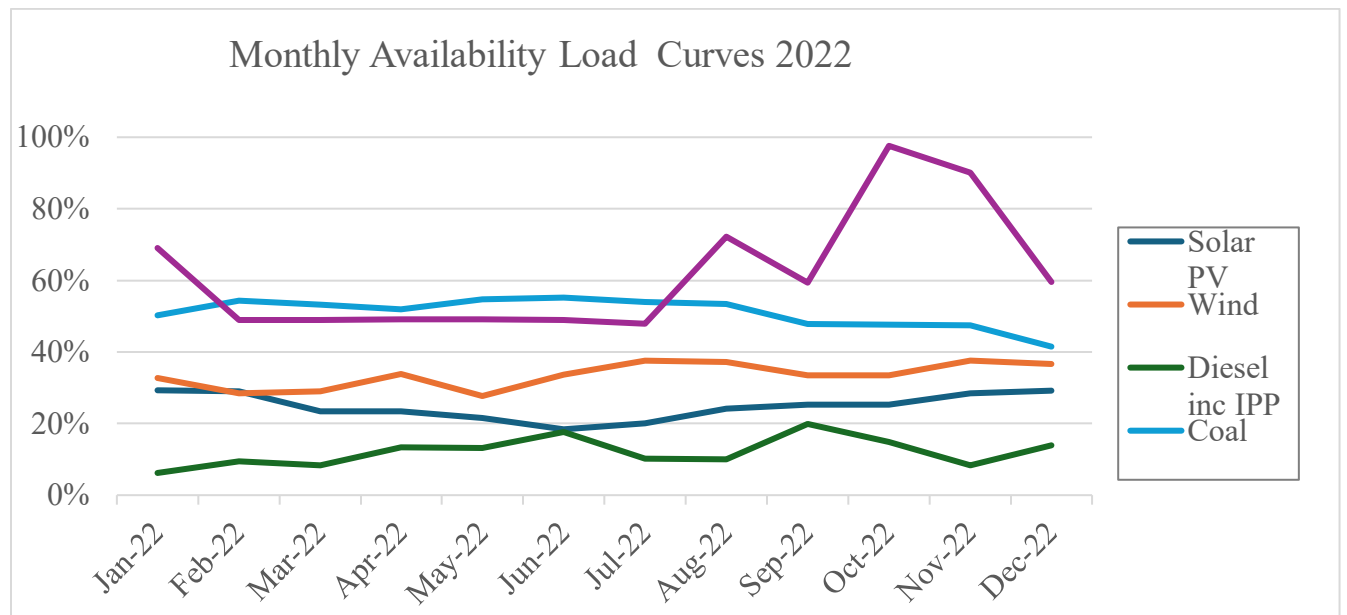
$$\rightarrow \text{Losses} = (\text{Total Supply} - \text{Total Sales}) / \text{Total Supply}$$

4. Other Important information:

a. Supply Curves

As the availability of plants differs on a daily and weekly basis, we found representative monthly capacity factors (CSIR, 2022) which were input as system availability load curves in LEAP. While this is a potential input, they are not currently being applied to the fuel sources below.

Figure 3: Monthly Load Curves as per CSIR



b. Future Generation

Future projections of electricity generation from each source were assumed under the “Running Cost” dispatch rule, which selects a generation source based on running cost (fuel cost). The future generation depends on capacity, including future capacity, capacity factor / maximum availability %, and the projected demand. LEAP uses all these factors and projects the future generation required to meet demand.

5. Supply figures for Scenarios:

a. Business as Usual (BAU)

i. Future Eskom Generation & IPP

We utilized South Africa’s Integrated Resource Plan (IRP 2023) to understand Eskom’s plans until 2050. We added and retired plants as per the tables and figures from the report included in the appendix. The calculations are detailed in our Energy Parameters Excel file: ‘Eskom + CT IPP Supply forecast’ sheet. We utilized these tables to find capacity after adding and retiring respective plants. For coal plants specifically, we found individual Eskom coal plants and retired them exactly as the plan shows. We gathered data on the Eskom coal plants that were missing in the IRP from Eskom’s website (Coal fired power stations - Eskom, 2021). After determining the projected annual change in Eskom’s supply until 2050, we calculated the percentage increase or decrease in capacity for each year and then applied those percentages to our 2023 Eskom Generation and IPP values.

ii. Future Cape Town Generation & IPP

We utilized Cape Town’s state of energy and carbon 2021 report to model the city’s plans for direct energy procurements until 2050, as shown in Figure 4, which summarizes the city’s plan to increase purchases from solar and wind IPPs:

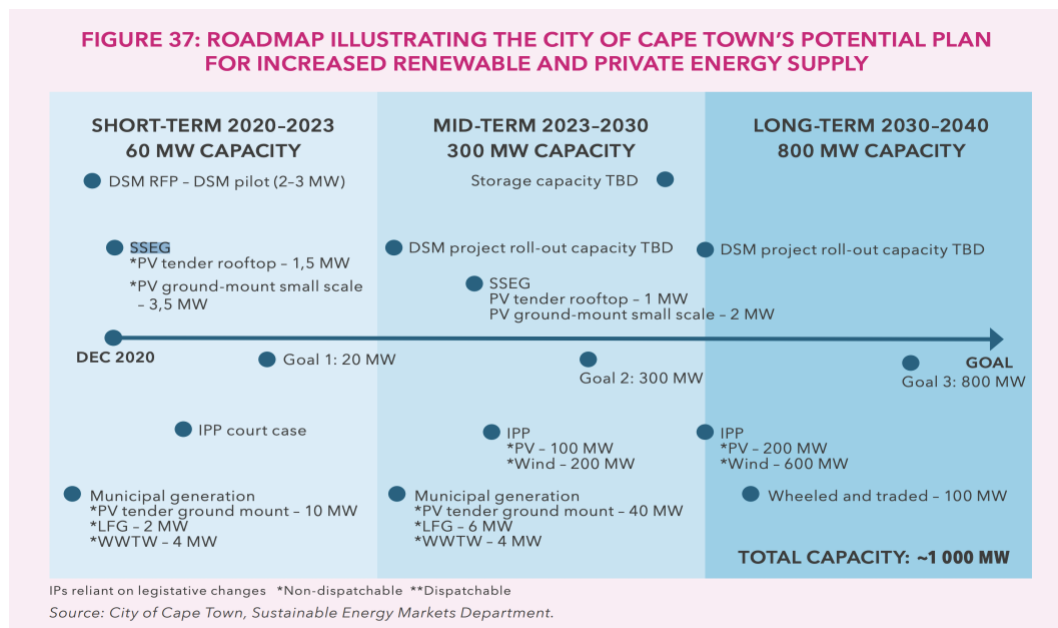


Figure 4: Future Cape Town Generation & IPP-procured capacity

The calculations are detailed in our Energy Parameters Excel file: ‘Eskom + CT IPP Supply forecast’ sheet. However, the report mentioned nothing about Cape Town’s plans for its existing owned plants: AGT and RGT. Because these plants are quite old and had been previously decommissioned, we assumed that they would be retired by 2035.

iii. Future SSEG Feed-in

The SSEG feed-in was calculated based on a few assumptions from the City of Cape Town records of installed capacity, the 2023 IRP demand model estimations of Onsite PV, and the RES4Africa report on the techno-economic potential for rooftop PV in Cape Town (Merven, 2023; Riva et al., 2024). For our Business-As-Usual (BAU) Scenario, we extrapolated current installed capacities using the final growth rate of 8% specified for 2030 in the 2023 IRP.

According to the RES4Africa report, the current techno-economic potential of rooftop solar capacity in Cape Town is 4,186 MW, which we used to set a 2050 target for Cape Town for our City of Cape Town Renewable Energy Expansion (CCTREE) Scenario described in the subsequent section. Using this figure and our 2023 capacity value for feed-in, we calculated a long-term compounded annual growth rate as follows:

➔ Growth Rate: $RRI(nper, PV, FV)$

- $nper$ = nth period, so from 2023 to 2050, the year 2050 will be 27th year.
- PV = this is 2023 capacity value for SSEG feed-in
- FV = this is the RES4Africa 2050 capacity figure for SSEG feed-in

Model Overview

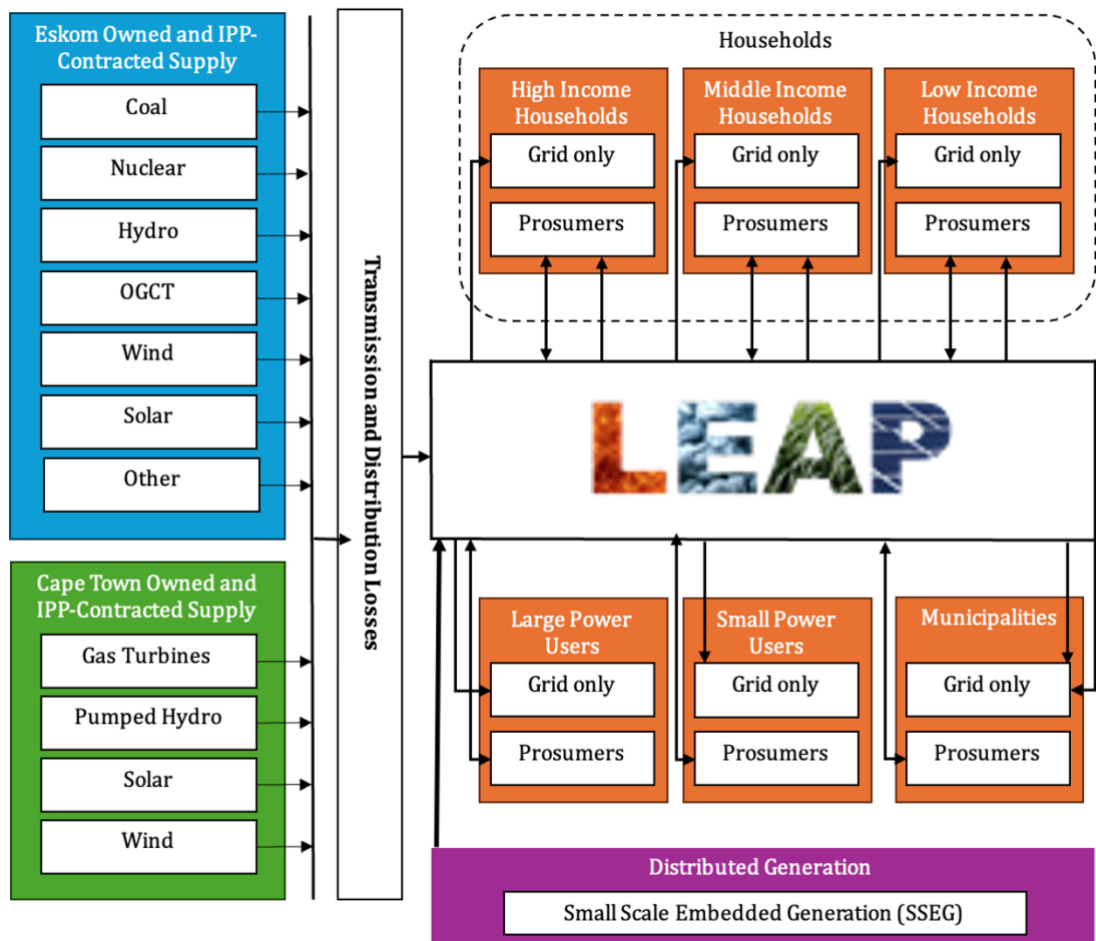


Figure 5: Model Overview

Scenarios

We leveraged LEAP's scenario analysis capabilities to assess the impacts of different policies and penetration of Small-scale embedded generation (SSEG) and its impact on demand on the Cape Town energy system as follows:

- a. **Business-As-Usual (BAU) Scenario:** This scenario provides a baseline assessment of how the Cape Town energy system might fare under current energy supply and demand assumptions based on the 2023 South African Integrated Resource Plan (IRP) reference case. The 2015 City of Cape Town future energy planning document includes a scenario built on 2013 IRP assumptions, which affirms the use of the 2023 IRP as appropriate for our model. The IRP reference case leverages the electricity demand forecast by the Energy Systems Research Group at the University of Cape Town alongside supply-side initiatives under construction by both the public and private sectors, either, which have a commercial operation date or are included in public procurement initiatives by the South African Department of Mineral Resources and Energy (Department of Mineral Resources and Energy, 2023). For the supply model, we based our future growth rate for SSEG in this scenario on the final projected growth rate of SSEG in 2030 of 8% as per the demand model used in the 2023 IRP (Merven, 2023). On the demand side, we account for this growth using values benchmarked from the *Embedded Solar PV Scenario* in Cape Town's 2015 update to its Energy Scenarios report, which projected 50% of high-income households would have SSEG, while commerce and industry would also have 15% of their electricity met by SSEG by 2040 (City of Cape Town & SEA, 2015). For our analysis, we extended the 50% target to middle-income households since the projected capacity with the extrapolated growth rate of 8% is sufficient to allow this, and we used the 15% target for SSEG consumption by LPUs, SPUs, and the Municipality by 2040.
- b. **City of Cape Town Renewable Energy Expansion (CCTREE) Scenario:** Relative to the BAU scenario, this scenario considers higher investment from the City of Cape Town in renewable energy (RE) capacity to reduce reliance on Eskom and mitigate load shedding. In this scenario, we consider the impact of increased SSEG capacity associated with potential policy initiatives to install more solar home systems in high- and middle-income households, alongside enhanced procurement of renewable energy by the City of Cape Town directly from IPPs as proposed in the City's 2050 Energy Strategy (City of Cape Town, n.d.-a). To estimate the growth rate for SSEG in this scenario, we used the total feasible economic capacity that could be installed based on the RES4Africa assessment of techno-economic potential and calculated the CAGR required to achieve that capacity by 2050 (Riva et al., 2024). To reconcile this growth in SSEG capacity on the demand side, we adopted the same demand-side SSEG adoption rates highlighted in the BAU scenario, accelerating the timeline to meet the targets to 2030 instead of 2040. For our analysis, we focused on the contribution of distributed SSEG in meeting energy demand in Cape Town and mitigating potential load-shedding impacts (Jeuland & Yoder, 2024).
- c. **Conservative Integrated Resource Plan (CIRP) Scenario:** Relative to the BAU scenario, this scenario represents the outcomes in Cape Town if South Africa fails to build new capacity at the rate of the 2023 IRP reference case in line with the more pessimistic *Firm Initiatives* scenario outlined in the IRP. The *Firm Initiatives* scenario considers only supply-side initiatives under construction and those ranked as a priority by the private

sector, providing a more pessimistic view of capacity additions relative to the IRP reference case (Department of Mineral Resources and Energy, 2023). Using data from the 2023 IRP, we included only *Firm Initiatives* in capacity additions of Eskom solar and wind between 2024 and 2030, and compared the impacts on load shedding, given SSEG capacity expansion equivalent to the BAU scenario.

For all scenarios, the modeling period is 2018 – 2050, using actual historical data for the period 2018 – 2023.

Results and Discussion

Business-As-Usual Scenario

Demand

Large Power Users make up most of the consumption annually, ranging from 39-43% depending on the year, differing due to growth rate assumptions. Household consumption varies between 35-37% of overall demand, while Small Power User consumption comprises between 16-20% of overall demand in the projected scenario. The Municipal sector makes up the least amount of annual demand at approximately ~4% of annual demand.

For the BAU scenario, we find that total electricity demand in Cape Town increased from 7,985.9 GWh in 2023 to 9,643.9 GWh in 2050. We note historical fluctuations in demand, which is a function of our use of electricity sales data for our analysis. Since the energy sales volumes by the City of Cape Town that we based our demand on dropped year on year from 2018 to 2020, subsequently increasing until 2022 before dropping again in 2023, our historical trend follows the same pattern. Future energy demand is projected to increase more steadily since it is based on a standardized CAGR.

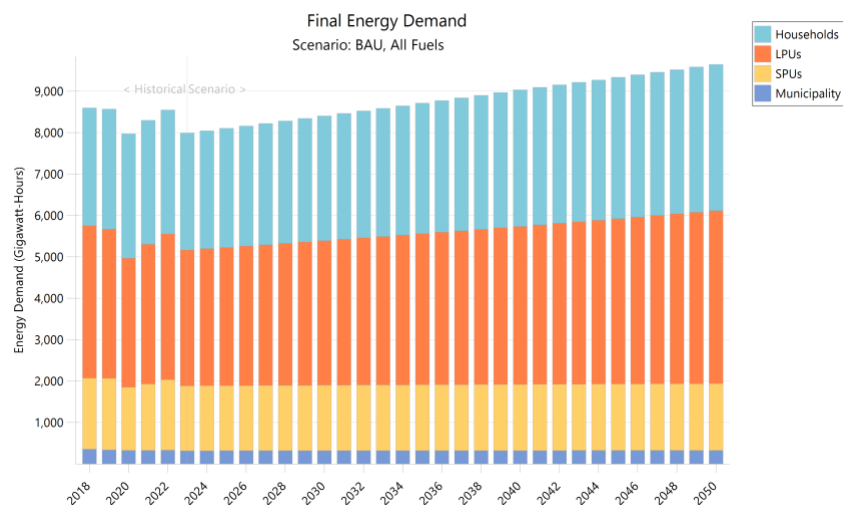


Figure 6: Electricity demand projection from 2018-2050 under BAU scenario

While we tried to benchmark our projected electricity demand with the recorded values from the Cape Town 2021 State of Energy and Carbon Report to validate our analysis, we noted that our projections for 2018-2023 annual household consumption were lower than previous estimations

(11,776 GWh in 2021-2022). This discrepancy is largely due to using electricity sales data rather than actual measured household energy consumption values. It is also difficult to compare directly as the reported values in the SOEC report include additional sectors, such as Transport and Agriculture, which were not represented in our LEAP model.

Supply

Eskom and Cape Town IPPs (Grid Supply)

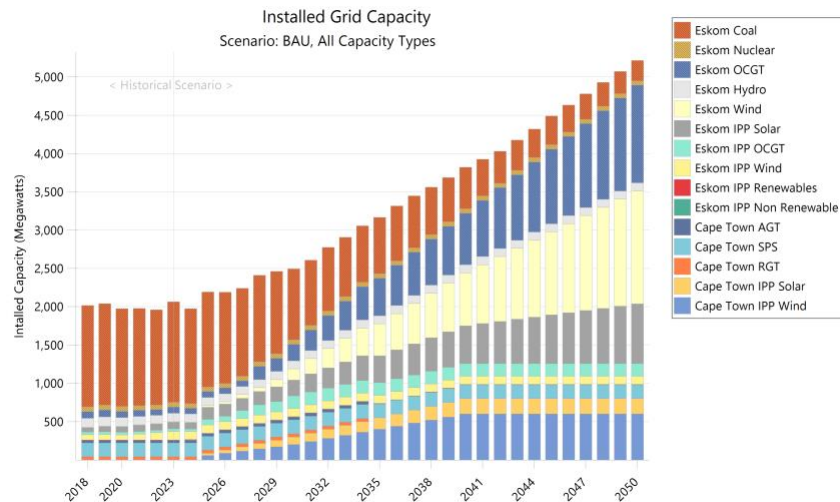


Figure 7: Installed Grid Capacity (2018 – 2050)

Based on our supply projections, the installed capacity of generation available to the City of Cape Town is projected to increase from an estimated 2,062.9 MW in 2023 to 5,216.0 MW in 2050. The fluctuations observed in the historical capacity occur because we calculated the proportion of Eskom’s installed capacity available to the City of Cape Town as the fraction of Cape Town’s peak demand to Eskom’s peak demand. Hence, changes in the relative demand peaks for the City of Cape Town and Eskom impacted how much capacity we estimated as historically available. Notably, we incorporated the planned coal capacity retirements as per the shutdown plan highlighted in the 2023 IRP. As a result, the proportion of installed coal capacity decreases from 63.5% in 2023 to 5.1% in 2050. The estimated proportions of Eskom-owned wind capacity and Eskom IPP-procured solar capacity are also projected to increase from 0.2% and 4.4% of installed capacity, respectively, in 2023 to 28.3% and 14.9% in 2050.

Distributed Generation

The increasing capacity of SSEG, even in a BAU scenario, is observable in the figure below. Since we assumed that distributed generation is dispatched at full capacity in our model, the energy generation from SSEG is projected to follow a similar trend to that observed in the capacity growth.

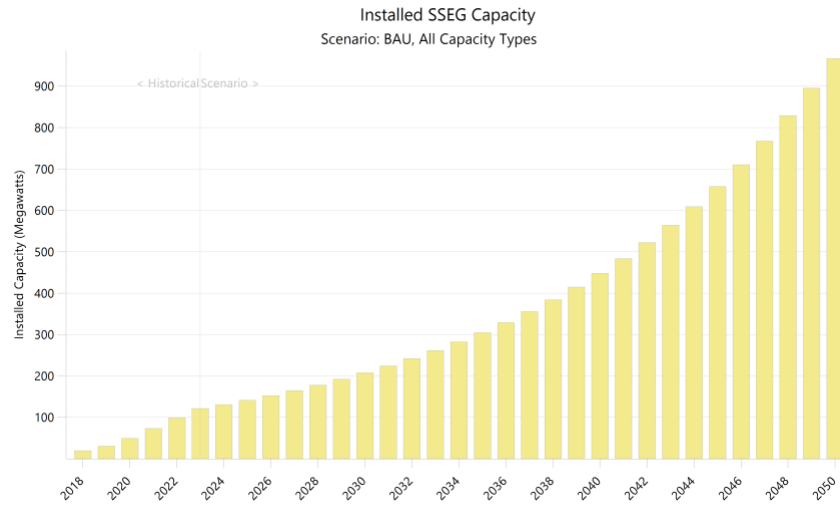


Figure 8: Installed SSEG Capacity (2018 – 2050)

Electricity Generation

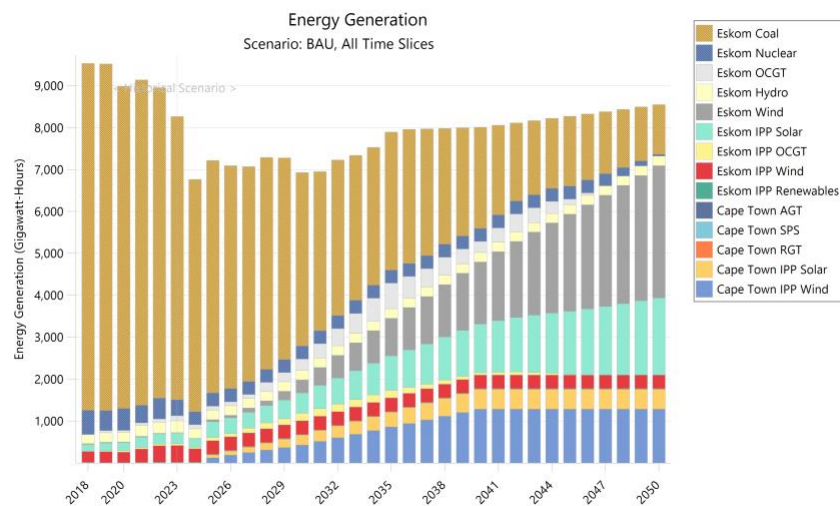


Figure 9: Annual Energy Generation (2018 – 2050)

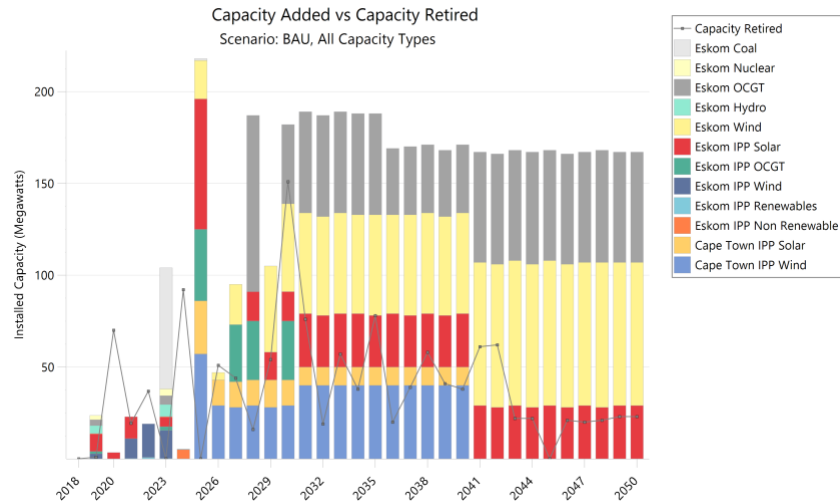


Figure 10: Capacity Added vs Capacity Retired (2018 – 2050)

From our BAU modeling results, we find that if the planned coal shutdown plan highlighted in the 2023 IRP is followed through, coal generation drops significantly, as highlighted by the drop observed between 2024 and 2035 (Figure 9). The drop in coal generation is because of planned retirements amidst growing demand and relatively low proportions of firm capacity (Figure 10). This results in unmet requirements as defined in LEAP, which can be interpreted as load shedding (Figure 11).

Load Shedding (Unmet Requirements)

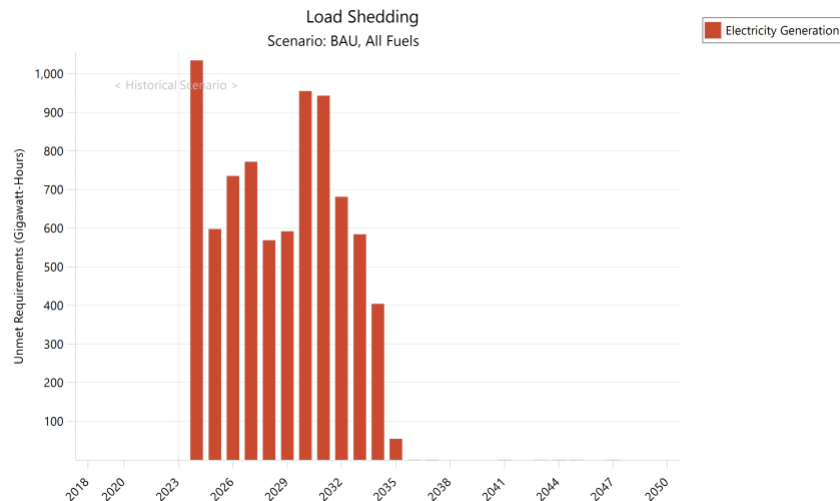


Figure 11: Projected Load Shedding

CCTREE Scenario

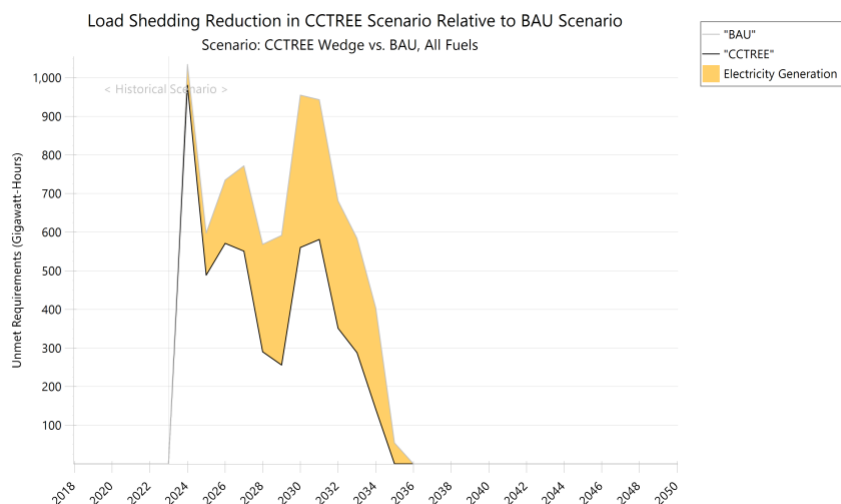


Figure 12: Load Shedding in CCTREE Scenario relative to BAU Scenario

From the CCTREE scenario, we find that increasing the annual growth rate of SSEG capacity from 8% to 14%, which matches the growth rate required to maximize economically viable installations on existing roof spaces by 2050, would significantly mitigate load shedding by a total of 2,861.5 GWh between 2024 and 2035. In this scenario, the proportion of distributed generation meeting final energy demand increases significantly as observed in the Sankey diagrams in Figures 13 and 14. Notably, in the Sankey diagram in Figure 14, highlighting energy flows in 2050, there is significant excess SSEG generation observed in the *Renewables* branch, partly due to a main limitation of our model that requires manual reconciliation of SSEG generation and demand.

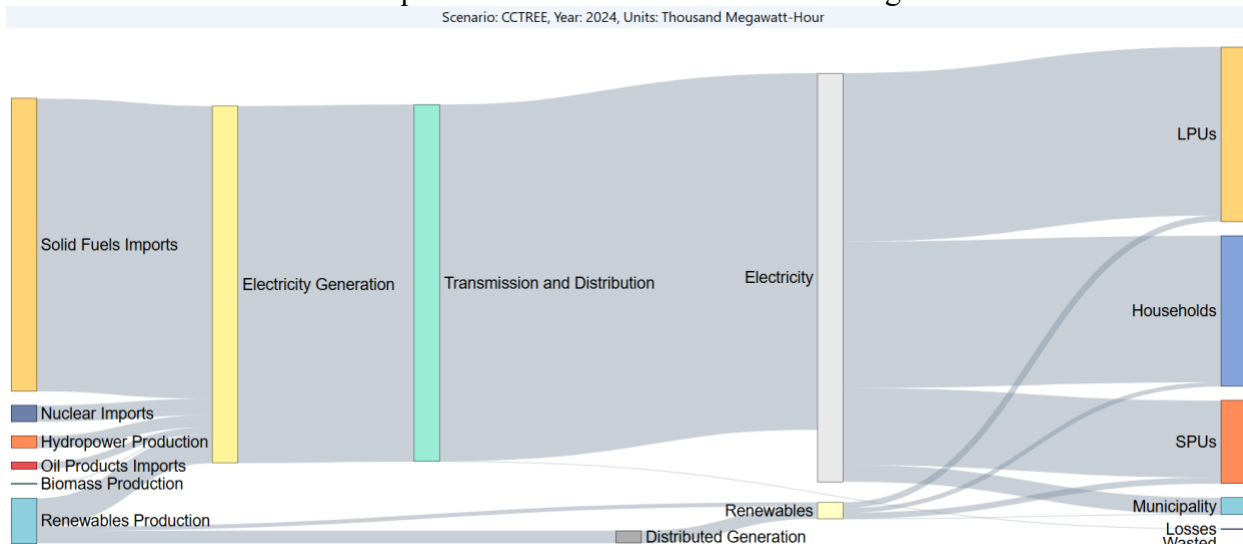


Figure 13: Energy Flows in the First Simulation Year (2024) in the CCTREE Scenario

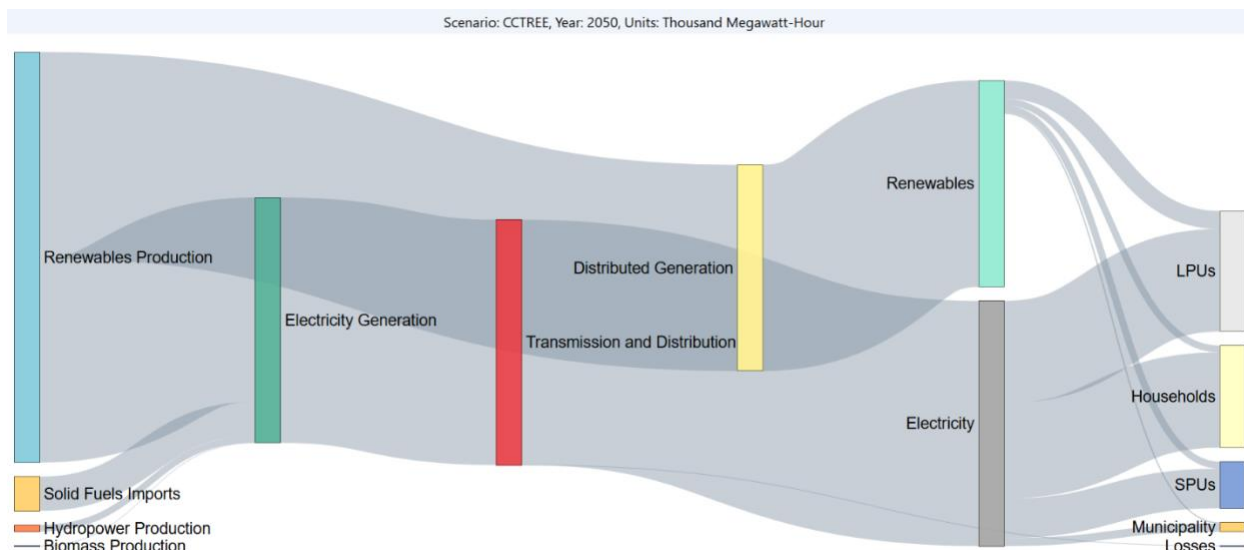


Figure 14: Projected Energy Flows in the Final Simulation Year (2050) in the CCTREE Scenario

CIRP Scenario

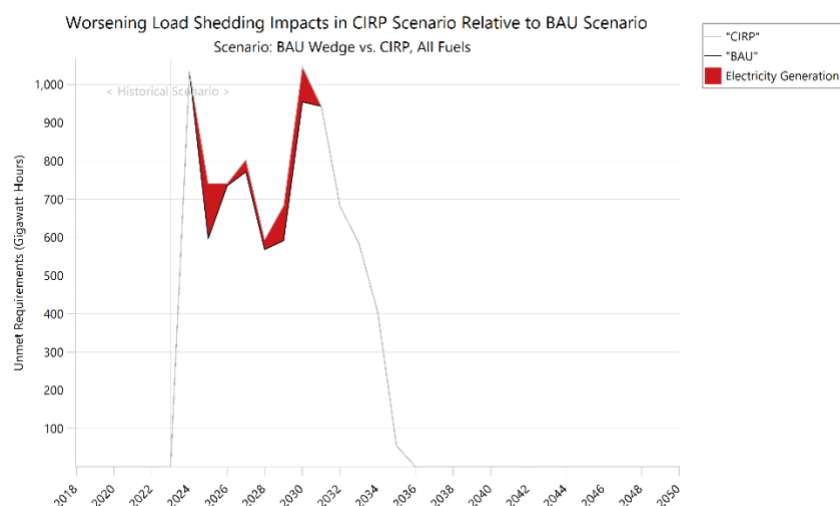


Figure 15: Load Shedding in CIRP Scenario relative to BAU Scenario

From our CIRP scenario, in which the only difference with the BAU scenario is a reduction in the proportion of solar and wind projects that come online between 2024 and 2030 (aligned with the *Firm Initiatives* scenario within the IRP), we find that load shedding impacts worsen, increasing by 392.6 GWh in that period. The results from this scenario highlight how delays in planned projects coming online may significantly worsen load shedding impacts, which could be mitigated through SSEG, as seen in the CCTREE scenario results.

Table 17: Summary of Scenario Results

	Projected Demand (GWh)			Final Energy Demand from SSEG (GWh, %)			Load Shedding (Unmet Requirements, GWh)
Scenario	2030	2040	2050	2030	2040	2050	2030
Business-As-Usual (BAU) Scenario	8,398	9,031	9,644	524 6%	1032 11%	1104 11%	955
City of Cape Town Renewable Energy Expansion (CCTREE) Scenario	8,466	9,031	9,644	986 12%	1080 12%	1153 12%	561
Conservative Integrated Resource Plan (CIRP) Scenario	8,398	9,031	9,644	524 6%	1032 11%	1104 11%	1047

The summarized results from all the scenarios are displayed in Table 17. As observed in the load shedding outcomes, we find the CCTREE scenario results in the biggest mitigation effect, with SSEG meeting 12% of projected electricity demand by 2030. While we used the same demand projection across all scenarios, the projected demand in 2030 in the CCTREE scenario is higher due to a higher penetration of households with SSEG. In our model, households with SSEG have slightly higher energy consumption, with the extra demand met by SSEG. Hence, the projected demand appears slightly higher in the CCTREE scenario, where 50% of high-income and middle-income households adopt SSEG by 2030.

In Table 7, the share of final energy demand from SSEG is observed to be flat from 2030 in the CCTREE scenario, while flat from 2040 in the BAU and CIRP scenarios. This occurs because in the CCTREE scenario, we set the target for SSEG adoption to 50% of middle- and high-income households, and 15% of LPU, SPU & Municipality consumption from SSEG by 2030. Once the target is met, all subsequent years have the same % since SSEG might be increasing on the supply side, but we don't see any further growth on the demand side apart from the IRP-based aggregate demand growth we specified. Similarly, in the BAU and CIRP scenarios, the same ratios apply for 2040 since the SSEG target is set for 2040, and we observe a fixed SSEG penetration rate after the target year.

Discussion And Policy Implications

Integrating Small Scale Embedded Generation (SSEG) as a grid supply option for Cape Town presents both challenges and opportunities. SSEG may reduce periodic loadshedding, resulting in a more resilient energy system. However, SSEG expansion also introduces significant implications for municipal revenue, customer equity, and long-term sustainability.

Drawing from the 2019 study by Sustainable Energy Africa, municipalities across South Africa have shown a wide range of tariff approaches in response to SSEG uptake, highlighting that tariff design is critical to managing revenue impacts. While many municipalities fear revenue loss due

to reduced electricity sales from residential customers, this can be mitigated with carefully structured SSEG tariffs. Effective strategies typically involve a combination of fixed monthly charges to cover network costs, consumption-based energy charges, and export tariffs that fairly compensate customers for electricity fed into the grid. For Cape Town, integrating SSEG could lead to variable revenue outcomes depending on the tariff model adopted, ranging from revenue gains (up to +7%) to notable losses. Moreover, the financial attractiveness of SSEG for customers measured through payback periods depends heavily on tariff balance, with the most equitable models offering 8–12-year paybacks. Commercial customers are less likely to impact municipal revenue negatively, as they already pay fixed and demand charges, unlike residential users. To ensure Cape Town maintains financial stability while enabling sustainable energy access, it should adopt best practices, including fixed charges, export tariffs that reflect avoided bulk electricity purchase costs, and time-of-use tariffs to better align pricing with actual system costs. Through well-structured tariffs, incorporating SSEG can have a neutral revenue impact for municipalities due to reduced reliance on power purchases, minimized technical losses, and cheaper sold electricity. Finally, Cape Town must anticipate and plan for the long-term erosion of the traditional 15% revenue surplus from electricity sales, shifting towards a more service-oriented revenue model that supports both municipal and customer resilience in the energy transition (Africa, 2019; Dippenaar et al., 2019).

A long-term outlook may also involve incorporating residential time-of-use tariffs as bi-directional meters become more prevalent and cost-effective. Cape Town is currently exploring net-billing for licensed distributors, which allows more cash benefits for prosumers at a wholesale rate; however, a net metering structure, incorporating time-of-use tariffs that are real-time, may be more attractive for customers and grid operators (National Energy Regulator of South Africa (NERSA), 2024; National Renewable Energy Laboratory (NREL), 2025). Customers will be properly credited for excess energy and compensated more similarly to Eskom's energy purchases, minimizing potential economic losses to the utility. While there may be a better long-term outlook, there are more technical challenges as smart meters must be implemented. The evolution of battery storage technologies may also help shape Cape Town's future tariff structure. Time-of-use tariffs may incentivize customers to stay connected to the larger grid to maximize potential profit and utility with energy storage technologies (Dippenaar et al., 2019).

Pakistan's experience with the rapid rise of SSEG offers a cautionary example for Cape Town and other cities considering widespread SSEG integration. As Pakistan undergoes a solar boom, the grid has become more financially strained, with systemic vulnerabilities exposed due to structural and contractual rigidities. Most critically, Pakistan's power sector operates under "take-or-pay" contracts with Independent Power Producers (IPPs), obligating utilities to make capacity payments regardless of actual demand. As more high-consumption commercial and industrial users switch to SSEG, grid demand declines, but the fixed payment obligations persist, creating a situation where most of the power purchase cost is locked in as capacity charges. This triggers a form of the "utility death spiral" as fewer users remain on the grid, leading to rising tariffs for the residential customers who are the least financially equipped to bear the costs. In addition, the breakdown of cross-subsidies, where commercial users previously helped offset residential electricity costs, further weakens utility revenue and grid equity. Technical challenges such as reverse power flow, voltage fluctuations, and daily "duck curve" effects also strain a grid infrastructure designed for one-directional flow. Meanwhile, poor regulatory oversight and substandard imports have led to

reliability concerns. Aggressive solar uptake may jeopardize long-term planning by rendering large-scale generation investments at risk of becoming stranded (Amjad, et al., 2024).

These dynamics highlight that while SSEG offers a path to clean energy access, it must be accompanied by careful tariff restructuring, policy reforms, and robust grid modernization to ensure that the transition is economically and technically sustainable. These lessons are highly relevant to Cape Town's own evolving energy landscape, which looks to expand SSEG integration. Reforming the city's tariff structure with a more reflexive and dynamic strategy will not only benefit residential consumers and municipalities but is also key to mitigating load-shedding impacts, a critical priority for Cape Town's energy strategy through 2050.

Future Work and Current Limitations

The incoming energy modeling team will have an established BAU scenario model representing the main contributors to the City of Cape Town's electricity sector, alongside preliminary models for the CCTREE and CIRP scenarios. Future work can improve distributed energy production and consumption estimations from the machine learning workstream's aerial imagery analysis. These estimates will then be able to provide sensitivity analyses to the BAU, incorporating higher demand growth due to other drivers such as EV and data center deployment. Additionally, the future team will be able to build upon and undertake additional sensitivity analysis of the CCTREE and CIRP scenarios explained above.

For our proposed scenarios, the impact of SSEG integration on Cape Town's total energy system cost and on household energy costs should be explored through an energy affordability assessment. With the Affordability Analysis tool in LEAP, the costs to different consumer categories in each scenario and the impacts on the respective consumer incomes can be determined (Stockholm Environment Institute, n.d.-a). Understanding utility profitability using the outputs of the LEAP model, alongside publicly available data on Cape Town's tariff structure and Eskom's tariff for local authorities, may also be of interest (City of Cape Town, n.d.-b; Eskom, n.d.). Household consumer and utility cost analysis can inform battery and SSEG integration. As battery costs decrease, this will be especially important in future model years (2030-2050).

During the modeling process, our team encountered and noted several limitations, some related to data availability and others due to limitations of the modeling software. In our model, distributed generation is modeled on the supply side as cumulative installed capacity and on the demand side as the proportion of households with SSEG or the fraction of energy from LPUs, SPUs, and the Municipality supplied by SSEG. Hence, modeling future scenarios requires reconciling the expected capacity in future years based on the growth rate of installed capacity while also increasing the share of load met by SSEG on the demand side, either by increasing the share of households with SSEG or by increasing the share of consumption from SSEG by LPUs, SPUs, and the Municipality. This manual process lends itself to error and will need to be refined by future teams to improve model accuracy, particularly when inputting installed capacity from the geotagged datasets of rooftop solar capacity from the machine learning workstream. The SSEG growth is an important assumption to analyze further, as this is essential to Cape Town's energy strategy. Future teams may examine other strategies to incorporate distributed energy resources on the supply side. This may involve looking at outsourcing LEAP estimates to other software.

For simplicity in our model, we also assume that households with SSEG consume more energy than households without, since our modeling approach adds extra consumption from SSEG to the average grid consumption for households in an income category. The insights from the econometrics workstream on household energy consumption could be used to refine the analysis further, since accurate consumption figures will be integral for the affordability analysis.

Another important limitation of LEAP is integrating battery storage. Since we are modeling an increase in distributed generation for Cape Town, the analysis is incomplete without battery storage, which often accompanies these solar photovoltaic systems and is necessary for tackling the intermittent nature of solar generation. However, the current model does not incorporate storage since battery storage can only be modeled using LEAP's optimization calculations (Heaps, 2022).

Some of the data-related limitations we encountered arose from insufficient publicly available data. Electricity sales data provided to us was only available until June 2023, hence, we extrapolated for the missing months in 2023. Furthermore, since Eskom's integrated annual reports are based on a financial year that runs from July to June, there is a temporal difference in some of the data we collected from Eskom's reports and some of the other data used in our model, which is based on a calendar year. Lastly, we model a fixed ratio of households across the income groups from 2024 – 2050, and a uniform increase in consumption within the groups due to limited data. However, realistically, there would be changes in the proportion of households across the income groups due to economic growth and different energy consumption growth rates within those groups, a factor that could be accounted for by future teams.

Conclusions

Cape Town is currently in an energy crisis. While the need for a renewable, more decentralized energy system is well known, the transition brings up overarching technical and social challenges. Cape Town has already begun to see an increase in renewable energy generation, especially within the residential sector, from the adoption of small-scale embedded generation (SSEG). Based on current Integrated Resource Plan developments, the city is projected to reach 9,644 GWh of demand by 2050. This is matched with an increasingly smaller share of coal due to coal plants retiring. From our analysis, we find that enhanced SSEG adoption can significantly contribute to load shedding mitigation. We also note that SSEG can play an especially crucial role in load shedding mitigation if there are delays in getting new projects online that would exacerbate load shedding impacts. However, we estimate that load shedding may still be expected to occur through 2035.

Cape Town must consider reliable energy planning alternatives to mitigate load shedding for a more resilient strategy, as increased SSEG penetration may be harmful without intervention. While SSEG growth is expected to provide reduced electricity tariffs to consumers, redesigning the current tariff structure may be needed to remain beneficial for municipalities and utility providers.

Disclosure statement

AI tools were used to slightly modify a few sentences in the Introduction section only. All team members contributed equally to the scope of work. Team members individually learned how to use the LEAP platform using the publicly available training material and met to discuss our learnings and challenges in a weekly team meeting on Wednesday. Subsequently, we collectively built the final energy model presented in our final presentation and in this report in a joint working session.

Appendix

Table 18: Demand modeling breakdown in LEAP

Sector	Sub-sector	Sub sub-sector	Type of Connection	Energy Intensity
Residential Sector (Number of Households)	Electrified (% share of Households)	Low Income (% share of Households)	Grid only	- Existing energy consumption (kWh/Household) - Projected energy consumption (kWh/Household)
			Grid and SHS	- Existing energy consumption (kWh/Household) - Projected energy consumption (kWh/Household)
		Medium Income (% share of Households)	Grid only	- Existing energy consumption (kWh/Household) - Projected energy consumption (kWh/Household)
			Grid and SHS	- Existing energy consumption (kWh/Household) - Projected energy consumption (kWh/Household)

Sector	Sub-sector	Sub sub-sector	Type of Connection	Energy Intensity
		High Income (% share of Households)	Grid only	- Existing energy consumption (kWh/Household) - Projected energy consumption (kWh/Household)
			Grid and SHS	- Existing energy consumption (kWh/Household) - Projected energy consumption (kWh/Household)
	Non-Electrified (% share of Households)	Low Income (% share of Households)	Grid only	- Existing energy consumption (kWh/Household) - Projected energy consumption (kWh/Household)
			Grid and SHS	- Existing energy consumption (kWh/Household) - Projected energy consumption (kWh/Household)
		Medium Income (% share of Households)	Grid only	- Existing energy consumption (kWh/Household) - Projected energy consumption (kWh/Household)
			Grid and SHS	Existing energy consumption (kWh/Household)

Sector	Sub-sector	Sub sub-sector	Type of Connection	Energy Intensity
				Projected energy consumption (kWh/Household)
		High Income (% share of Households)	Grid only	- Existing energy consumption (kWh/Household) - Projected energy consumption (kWh/Household)
			Grid and SHS	- Existing energy consumption (kWh/Household) - Projected energy consumption (kWh/Household)
Small Power Users	N/A	N/A	Grid only	- Existing energy consumption (kWh) - Projected energy consumption (kWh)
			Grid and SSEG	- Existing energy consumption (kWh) - Projected energy consumption (kWh)
Large Power Users	N/A	N/A	Grid only	- Existing energy consumption (kWh) - Projected energy consumption (kWh)

Sector	Sub-sector	Sub sub-sector	Type of Connection	Energy Intensity
			Grid and SSEG	- Existing energy consumption (kWh) - Projected energy consumption (kWh)
Municipal	N/A	N/A	Grid only	- Existing energy consumption (kWh) - Projected energy consumption (kWh)
			Grid and SSEG	- Existing energy consumption (kWh) - Projected energy consumption (kWh)

Table 19: Energy Supply Fuel Types

Source	Description
Coal	The majority of Eskom's generation capacity serves the base load.
Natural Gas	A smaller portion of Eskom's generation. Used primarily in gas turbines for peak load management.
Nuclear	The Koeberg Nuclear Power Station, located near Cape Town, Africa's only operational nuclear plant (Kincer & Lovering, 2023).
Wind	Eskom integrates wind energy from its 100 MW Sere Wind Farm and independent power producers (IPPs)
Solar	Contributions from utility-scale solar farms and distributed solar systems connected to Eskom's grid.
Hydro	Limited hydroelectric generation is part of Eskom's portfolio, primarily from "two conventional hydroelectric power stations, three hydro pumped storage schemes and four non-dispatchable mini hydro stations."
(Eskom Generation Division, 2024)	

Table 20: Energy Modeling Scenarios

Scenario	Demand-side assumptions	Supply-side assumptions	FUTURE WORK: Proposed Sensitivity analysis across scenarios
Business-As-Usual (BAU) Scenario	National electricity demand forecast in IRP 2023	Eskom supply based on the IRP 2023 reference case and Cape Town supply extrapolating current policies and trends in installation of SSEG	Load growth sensitivity: High and Low electricity demand growth relative to reference case
City of Cape Town Renewable Energy Expansion (CCTREE) Scenario		Eskom supply based on the IRP 2023 reference case and Cape Town supply based on a policy of relative higher investment from the City of Cape Town in renewable energy (RE) capacity from SSEG.	SSEG sensitivity: Modeling using data from official records vs. satellite imagery Tariff analysis: Sensitivity analysis of grid connection and/or feed-in tariffs
Conservative Integrated Resource Plan (CIRP) Scenario		Eskom supply based on the IRP 2023 <i>Firm Initiative</i> scenario and Cape Town supply extrapolating current policies and trends in installation of SSEG	

Table 21: Energy Supply and Peak Demand Base

	2018	2023
Peak Demand MW	1,765	2,000
Supply MW:		
Eskom	1,747	1,800
Cape Town - APT	36	36
Cape Town - RGT	42	42
Cape Town - SPS (pumped storage)	180	180
SSEG	19	121

Assumptions from the IRP used for Supply Modeling

➔ Emerging Plan from Horizon One (2024-2030)

Table 2: Emerging Plan from Horizon One Analysis

	Coal	Gas - IPP Programme	Gas - Eskom	Dispatchable Capacity	Nuclear	Hydro	Pumped Storage	CSP	Solar PV	Wind	Hydro IPP Programme	Distributed Generation	BESS - IPP Programme	BESS - Eskom	Unserved Energy (TWh)
Current Base (MW)	38 800	1 005	2 825	-	1 860	1 600	2732	500	2 287	3 443	-	5 000	-	20	
2024	720							100			150	900		199	13.06
2025	720	1 220							2 115	644	476	900	513	141	7.63
2026										140		900			7.66
2027		1 000								684		900	2 000	615	4.55
2028		1 000	3 000						300			900	615		0.22
2029									500	1 500		900			0.25
2030		1 000		1 376					500	1 500		900			0.27
Additional New Capacity (MW)	1 440	4 220	3 000	1 376				100	3 615	4 468	626	6 300	3 743	360	

Installed Capacity
 Capacity under construction
 Capacity procured
 New Capacity
 Distributed Generation Capacity for own use
 Unserved Energy, preferred as low as possible

Figure 16: Emerging Plan from Horizon One (2024-2030)

➔ Reference Energy Pathway new generation (2031-2050)

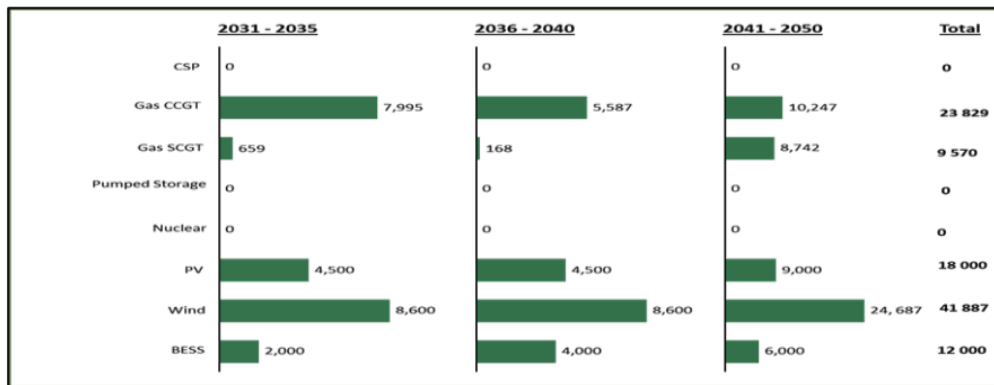


Figure 13: Reference Energy Pathway new generation capacity

Figure 17: Reference Energy Pathway new generation (2031-2050)

➔ Eskom coal fired station shutdown plan

	Arnot	Camden	Duvha	Grootvlei	Hendrina	Kendal	Komati	Kriel	Kusile	Lethabo	Majuba	Matimba	Matla	Medupi	Tutuka
U1	31-Mar-27	30-Aug-24	17-Aug-31	16-Aug-26	01-Apr-18	30-Sep-39	01-Apr-18	05-May-26	29-Aug-68	21-Dec-36	31-Mar-46	03-Dec-38	22-Aug-30	31-Jul-71	15-Mar-30
U2	31-Aug-26	30-Apr-25	30-Sep-31	21-Mar-26	10-Feb-25	19-Jun-41	01-Sep-18	13-May-27	30-Oct-70	10-Jul-37	31-Mar-47	03-Dec-38	29-Jun-31	30-May-69	10-Jan-30
U3	31-Jul-26	30-Nov-24	01-Dec-17	04-Sep-27	01-Jun-17	15-Dec-42	01-Sep-18	27-Jan-28	30-Aug-71	26-Mar-37	31-Mar-48	28-Sep-39	11-Dec-31	30-Oct-68	28-Jul-30
U4	31-Mar-27	31-Jan-23	30-Jun-33	01-Apr-18	31-Mar-25	30-Nov-42	01-Jun-20	21-Aug-29	30-Jun-72	02-Dec-38	31-Mar-49	29-Sep-40	15-Oct-32	30-Nov-67	22-Apr-30
U5	24-Nov-29	30-Nov-25	30-Mar-33	01-Apr-18	31-Dec-25	23-Dec-43	01-Jun-19	12-Mar-29	31-Dec-72	30-Jun-40	31-Mar-50	30-Sep-41	23-Aug-33	30-Apr-67	25-Sep-30
U6	31-Mar-29	31-Jul-23	21-Feb-34	01-Apr-18	13-Sep-25	09-Dec-44	01-Apr-18	16-Nov-30	30-Jun-73	27-Dec-41	31-Mar-51	30-Sep-42	20-Jul-34	31-Aug-65	05-Jun-30
U7		31-Jan-24			19-Mar-24		01-Sep-19								
U8		31-Jul-25			01-Jun-19		01-Apr-18								
U9					01-Oct-19		31-Oct-22								
U10					21-Mar-23										

Figure 18: Eskom coal fired station shutdown plan

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